Understanding Diabetes Progression

Data Analytics and Statistics with Applications — Final Report

QUESTION 1. UNDERSTAND AND PREPARE THE DATA

- **Row Representation:** Each row in the dataset represents a single patient diagnosed with diabetes. The dataset tracks a total cohort of 442 individuals across 11 distinct variables.
- **Variable Designations:** The primary outcome variable is progression (originally named 'y'), a quantitative measure of disease acceleration tracked one year after the baseline. The remaining 10 variables serve as predictors.
- **Nature of Data:** Ten variables are numerical/continuous (including age, bmi, blood pressure, and six blood serum panels labeled s1-s6). The variable sex represents a clear categorical attribute and has been explicitly converted into a factor.
- **Data Quality Audit:** The dataset contains 0 missing values (NAs) and 0 duplicate entries, providing an optimally complete foundation for modeling.
- **Responsibility:** Health-related data demands highly responsible interpretation because associations do not imply causation. Biased or misleading models can result in flawed medical guidelines or unnecessary patient anxiety.

QUESTION 2. EXPLORE THE DATA

- **Progression Metrics:** Displays a mean of 152.1 and a median of 140.5 (SD = 77.1). The distribution is moderately right-skewed, showing a dense cluster of stable patients and a long tail of rapid acceleration cases.
- **BMI Metrics:** Displays a mean of 26.38 and a median of 25.7 (SD = 4.42). This baseline establishes that the monitored group is generally overweight (BMI > 25).
- **Age Distribution:** Displays a mean of 48.5 years and a median of 50.0 years (SD = 13.1), reflecting a mature adult cohort.
- **Anomalies & Outliers:** No anomalous data spikes or recording errors are present. A maximum BMI value of 42.2 represents severe obesity but remains entirely plausible.

QUESTION 3. INVESTIGATE RELATIONSHIPS

- **Strongest Core Correlations:** Linear Pearson checks indicate that bmi (r = 0.59), s5 (r = 0.57), and bp (r = 0.44) display the strongest direct relationships with progression.
- **Redundancy and Overlap:** Massive multi-collinearity exists within the blood serum metrics, specifically between s1 and s2 (r = 0.90) and between s3 and s4 (r = -0.74). Including all serum elements concurrently would severely destabilize standard regression estimates.
- **Promising Modeling Predictors:** BMI, S5, and BP are identified as the most effective structural pillars for multiple regression because they provide high individual explanatory power with minimal initial cross-correlation.

QUESTION 4. BUILD AND INTERPRET REGRESSION MODELS

- **Simple Linear Model:** The fitted equation is *progression = -117.77 + 10.23 × bmi*. For each 1-unit increase in BMI, the baseline disease progression accelerates by 10.23 units ($p < 2e\text{-}16$). The intercept (-117.77) holds no real-world clinical validity since a BMI of 0 is impossible. This simple single predictor explains 34.4% of overall variation ($R^2 = 0.344$).
- **Multiple Regression Model:** The optimized model yields the following parameters:

Term	Estimate	Std. Error	t value	Pr(>|t|)
Intercept	-334.88	25.88	-12.94	< 2e-16
bmi	6.50	0.697	9.32	< 2e-16
bp	0.90	0.217	4.17	3.7e-05
s5	49.58	5.890	8.42	5.6e-16

- **Interpretation:** Holding all other clinical variables constant, a 1-unit increase in BMI yields a 6.50 unit increase in progression, while a 1-unit increase in serum marker s5 prompts a major 49.58 unit increase.
- **Performance Gain:** Expanding to the three-pillar model drives the explained variance up to 48.0% (Adjusted $R^2 = 0.477$), dropping the standard residual error from 62.5 down to 55.8.

QUESTION 5. EVALUATE THE ANALYSIS

- **Diagnostic Plot:** The Residuals vs. Fitted value mapping shows a stable, random scatter around the zero line without any severe funneling or non-linear patterns, satisfying homoscedasticity.
- **Multicollinearity:** Variance Inflation Factors (VIF) are exceptionally clean (BMI: 1.345, BP: 1.274, S5: 1.342). All scores fall far below standard critical thresholds, confirming model stability.
- **Significance vs. Practicality:** All three metrics are highly statistically significant ($p < 0.05$). However, BMI possesses distinct practical importance because it represents an actively modifiable behavioral target, unlike raw internal serum concentrations.

QUESTION 6. DRAW APPROPRIATE CONCLUSIONS

- **Observational Framework:** This tracking setup represents an observational framework, meaning no causal relationships can be definitively established.
- **Unmeasured Factors:** Key unmeasured confounding factors—such as precise patient dietary habits, exercise consistency, or strict genetic lineages—remain outside the active data framework.
- **Healthcare Recommendation:** Clinical management and resources should focus on targeted, preventative weight management and automated blood pressure monitoring systems to actively decelerate long-term diabetes progression trajectories.